

FIONA DARK

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Full stack software engineer with 1.5 years of experience building backend systems, real-time data pipelines, and distributed cloud-native applications. Skilled in C++, Python, and Java, with a focus on delivering reliable, production-ready code.

EDUCATION

Fordham University

September 2021 - May 2025

Bachelor of Science in Computer Science

- **Honors:** Graduated *summa cum laude*, Dean's List (Fall 2021 - Spring 2025), recipient of the Herbert W. Bomzer Award in Computer and Information Sciences
- **Relevant Coursework:** Data Structures, Java Development, Algorithms, Database Systems, Operating Systems, Data Communication & Networks, Cybersecurity, Data Mining, Theory of Computation, Mobile Application Programming

SKILLS

- **Programming Languages:** C++, Python, Java, JavaScript, Kotlin, SQL, HTML/CSS, Swift/SwiftUI, Groovy
- **Developer Tools & Platforms:** VS Code, IntelliJ IDEA, Xcode, Gradle, Maven, Docker, Jenkins, AWS, Datadog, Linux, Jira
- **Frameworks & Technologies:** Git, GitHub, GitHub Spec Kit, Spring Boot, JUnit, Apache Kafka, Apache Flink
- **Databases:** PostgreSQL, MySQL, Apache Cassandra
- **Development Practices:** Agile methodology, Test-driven development, CI/CD pipelines, Version control, AI-assisted development
- **Professional Skills:** Written and verbal communication, Time management, Project management, Team collaboration

EXPERIENCE

Arity

May 2025 - September 2025

Software Engineer Intern

Seattle, WA

- Engineered a high-volume Kafka pipeline processing millions of daily trip events, leveraging Apache Flink for real-time data stream processing and transformation.
- Built and refactored backend services with Java Spring Boot, integrating with Apache Flink and applying extensive unit testing to deliver robust, production-quality applications.
- Developed and containerized backend components using Docker and Apache Cassandra to ensure scalable and reliable deployment within a distributed cloud-native environment.

Fordham University

August 2024 - May 2025

Teaching Assistant

New York, NY

- Reviewed, tested, and debugged student code in C++ and Python, conducting informal code reviews and providing detailed feedback on implementation and style.
- Conducted office hours and pair programming sessions for over 70 students, guiding them through debugging and problem-solving strategies and reinforcing proficiency in C++ and Python.
- Compiled comprehensive study guides on C++, searching/sorting algorithms, and object-oriented programming principles.

US Environmental Protection Agency

May 2023 - August 2023

Software Engineer Intern

Seattle, WA

- Redesigned the SQL data model for a national climate change database, reducing redundancy and improving query efficiency; designed and built a UI prototype and incorporated user feedback.
- Built a national website of climate change resources for Native American tribes; prototype helped secure \$30,000 in federal funding for the project.

PROJECTS

The Little Library

- Developed a full-stack personal library management web application using Java, JavaScript, and Spring Boot, implementing RESTful APIs for book CRUD operations with user-specific data filtering and sorting.
- Integrated Firebase Authentication, Firestore NoSQL database, and the OpenLibrary API for secure user management and real-time book metadata retrieval.
- Deployed backend services on Railway and frontend on GitHub pages, architecting a scalable cloud-based application.

Augur: The Free Tarot Tool

- Developed a Java-based CLI Tarot reading application that integrates with a REST API to deliver an authentic divination experience.
- Engineered robust application architecture using Gradle build automation, Picocli framework for command-line interface, and Jackson for JSON processing, with comprehensive test coverage using JUnit 5 and Mockito.